

BANAK LAKSELV, Norway

WMO: 010590

Lat: 70.069N

Lon: 24.973E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.1	-21.2	-27.1	0.3	-23.9	-24.2	0.4	-21.0	18.1	-3.4	16.4	-4.1	2.6	180	0.826

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	22.9	15.2	20.6	14.3	18.3	13.3	16.2	21.3	15.0	19.4	13.8	17.6	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	18.4	12.8	9.2	16.7	11.8	8.6	15.6	45.4	21.5	41.9	19.4	38.8	17.6	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.9	11.2	DB	-26.7	27.1	3.3	3.0	-29.1	29.2	-31.0	31.0	-32.8	32.6	-35.2	34.8	(4)
(5)			WB	-27.0	18.3	3.2	1.7	-29.3	19.5	-31.1	20.5	-32.9	21.5	-35.2	22.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-8.4	-8.5	-4.9	0.0	5.1	9.5	13.0	11.4	7.9	2.1	-3.6	-5.6	(6)	
	DBStd	8.80	6.58	6.72	5.35	4.04	3.62	3.27	3.46	3.11	3.38	4.16	4.97	5.79	(7)	
	HDD10.0	3298	569	519	462	298	157	47	8	19	81	244	408	484	(8)	
	HDD18.3	6125	828	753	720	548	409	265	171	215	313	502	658	743	(9)	
	CDD10.0	221	0	0	0	0	6	32	101	63	17	1	0	0	(10)	
	CDD18.3	6	0	0	0	0	0	0	4	1	0	0	0	0	(11)	
	CDH23.3	59	0	0	0	0	0	2	44	12	0	0	0	0	(12)	
	CDH26.7	12	0	0	0	0	0	0	11	2	0	0	0	0	(13)	
(14) Wind	WSAvg	5.0	6.1	6.0	5.8	4.7	4.7	4.5	4.0	3.8	4.3	4.9	5.5	5.9	(14)	
(15) Precipitation	PrecAvg	467	32	28	28	25	31	45	57	58	48	47	32	36	(15)	
	PrecMax	573	62	51	52	46	84	79	133	154	94	88	69	72	(16)	
	PrecMin	321	8	6	5	7	7	8	11	27	12	20	6	13	(17)	
	PrecStd	65	16	12	12	10	18	20	31	28	18	19	16	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	6.2	7.2	11.6	20.2	23.4	28.1	25.3	19.0	12.8	7.8	7.7	(19)	
		MCWB	3.2	3.4	4.0	6.5	11.4	14.0	18.0	16.6	13.2	9.2	5.3	4.7	(20)	
	2%	DB	3.9	4.6	5.1	9.1	16.1	20.1	24.0	21.6	16.8	10.9	5.3	5.1	(21)	
		MCWB	1.7	2.1	2.3	5.2	9.7	12.7	16.1	15.0	12.3	7.9	3.3	2.5	(22)	
	5%	DB	1.9	2.3	3.3	7.0	13.1	17.4	21.4	19.1	14.9	9.2	3.8	3.2	(23)	
		MCWB	0.1	0.4	0.9	3.9	8.0	11.8	15.1	13.9	11.3	6.5	2.1	1.2	(24)	
	10%	DB	0.2	0.2	2.0	5.5	10.9	15.0	19.1	17.1	13.2	7.6	2.6	1.9	(25)	
		MCWB	-1.2	-1.2	-0.2	2.7	6.8	10.3	14.2	13.0	10.2	5.6	1.1	0.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.9	4.1	7.0	12.1	15.7	19.0	17.6	14.3	9.6	5.5	4.7	(27)	
		MCDB	5.7	6.2	7.2	10.6	18.8	20.4	25.8	22.9	17.8	12.2	7.4	7.3	(28)	
	2%	WB	1.6	2.1	2.5	5.4	9.8	13.7	17.0	15.7	12.8	8.2	3.4	2.7	(29)	
		MCDB	3.6	4.5	5.0	8.7	15.5	18.8	22.8	20.3	15.9	10.6	5.0	4.8	(30)	
	5%	WB	0.2	0.5	1.1	4.2	8.4	12.1	15.5	14.5	11.7	6.8	2.2	1.4	(31)	
		MCDB	1.7	2.2	3.1	6.9	12.5	16.6	20.4	18.4	14.5	8.6	3.5	3.2	(32)	
	10%	WB	-1.3	-1.1	-0.1	2.9	7.0	10.7	14.3	13.3	10.5	5.6	1.1	0.2	(33)	
		MCDB	0.1	0.2	1.8	5.1	10.4	14.5	18.5	16.6	12.9	7.4	2.4	1.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.6	7.0	6.8	6.7	6.6	7.4	7.7	6.5	4.7	5.5	6.3	(35)	
		MCDBR	7.4	7.4	5.9	7.7	10.9	11.2	12.2	11.9	9.1	5.8	5.8	6.6	(36)	
	5% WB	MCWBR	6.2	6.1	4.7	4.9	6.0	5.6	5.9	6.2	5.5	4.1	4.8	5.3	(37)	
		MCDWR	7.3	7.2	5.9	7.1	10.3	10.1	11.2	11.0	8.3	5.4	5.5	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.250	0.302	0.327	0.345	0.345	0.355	0.338	0.308	0.260	N/A	N/A	(40)	
		taud	N/A	2.031	2.102	2.159	2.296	2.449	2.456	2.483	2.520	2.183	N/A	N/A	(41)	
	Ebn at Noon		0	449	680	794	832	849	824	795	720	442	0	0	(42)	
		Edh at Noon	0	41	80	104	104	92	89	77	55	34	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.31	1.57	3.51	4.74	4.83	4.38	2.99	1.53	0.45	0.00	0.00	0.00	(44)	
	RadStd	0.00	0.03	0.10	0.20	0.42	0.39	0.46	0.28	0.14	0.05	0.00	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (22 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page